

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Medium

Question

The number of coins in a collection increased from **9** to **90**. What was the percent increase in the number of coins in this collection?

Answer

- A. **10%**
- B. **81%**
- C. **90%**
- D. **900%**

Correct Answer: D

Rationale

Choice D is correct. It's given that the number of coins in the collection increased from **9** to **90**. It follows that the number of coins in the collection increased by **90 — 9**, or **81**. Let ***x*%** represent the percentage that **81** is of **9**. The value of ***x*** can be found using the proportion $\frac{81}{9} = \frac{x}{100}$, or **9 = $\frac{x}{100}$** . Multiplying both sides of this equation by **100** yields **900 = *x***. Thus, when the number of coins in the collection increased from **9** to **90**, the percent increase was **900%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.